

ABHIJITH SIVAPRASADAN

Stockholm, Sweden | +46 76 969 2014 | abhijithsivaprasadan@gmail.com | linkedin.com/in/abhijith-sivaprasadan | github.com/abhijith-sivaprasadan

2 May 2026

Fredrik Bergström

Group Manager
Elomatic Oy
Espoo / Turku / Tampere, Finland

Re: Consulting Engineer CFD, Fluid Dynamics

Dear Fredrik Bergström,

I am applying for the **Consulting Engineer CFD, Fluid Dynamics** position at Elomatic. I am completing my M.Sc. in Sustainable Energy Engineering at KTH Royal Institute of Technology, and my strongest technical interest is in applying CFD, heat transfer, fluid physics, and structured numerical modelling to real engineering problems. Elomatic's work across energy, process industry, aerospace, and marine applications is very close to the kind of technical environment where I want to grow as a CFD professional.

My most relevant experience comes from my master thesis work at **Siemens Energy AB in Finspång**, where I worked as a Test Engineer / Master Thesis Student in the Fluid Dynamic Lab. I developed steady-state compressible CFD and conjugate heat-transfer models in **ANSYS Fluent** for high-temperature gas-turbine test-rig hardware. The work involved geometry preparation, meshing with inflation layers, boundary-condition definition, turbulence modelling using k-omega SST, convergence monitoring, post-processing, and comparison of geometry variants through engineering KPIs.

A major part of that work was ensuring that the simulations were not just visually convincing, but technically defensible. I conducted mesh-independence and near-wall validation studies, checked sensitivity of assumptions, documented model limitations, and connected simulation outputs to measurable quantities from the physical test rig. I also supported the experimental side by commissioning and verifying **NI-DAQ measurement chains** with thermocouples, pressure sensors, and LabVIEW acquisition workflows. This gave me a useful simulation-to-test mindset that I believe is important in consulting work, where results must be clear, traceable, and useful for customer decisions.

In addition to CFD, I bring a broader engineering and computational background. Through my KTH projects, I have worked with energy-system modelling, heating demand forecasting, district-heating optimisation, and thermal energy storage analysis using tools such as Python, MATLAB, IDA ICE, HOMER Pro, and SAM. Before my master's studies, I worked as a Backend Engineer at QBurst, which strengthened my ability to write maintainable code, debug systematically, document technical decisions, and work in collaborative project environments. These skills are directly useful for CFD post-processing, automation, technical computing, and consulting project delivery.

I am especially attracted to Elomatic because the role combines technical depth with variety: numerical modelling, customer-facing engineering, method development, and cross-sector fluid dynamics challenges. I am interested in continuing to develop within CFD for energy systems, process equipment, aerospace-related thermal-fluid problems, and future technologies such as hydrogen, marine energy systems, and sustainable industrial processes.

I should also mention honestly that I am fluent in English but not yet fluent in Finnish. I am motivated to learn Finnish and would be especially interested in projects where English can be used as the working language while I develop my Finnish over time. My preferred location would be **Espoo**, although I am open to discussing Turku or Tampere depending on project needs.

Thank you for considering my application. I would welcome the opportunity to discuss how my background in ANSYS Fluent CFD/CHT, thermal-fluid modelling, validation, test-rig instrumentation, Python-based analysis, and technical documentation can contribute to Elomatic's fluid dynamics team.

Sincerely,

Abhijith Sivaprasadan